Week 1- Profession

Ethics: Ethics is about human behavior — how a person acts in society or a company. Examples of ethical behavior: Saying "Sorry" when you make a mistake. Saying "Thanks" or appreciating others when they do something good
Law: Law is a set of rules enforced by an authority. Example: A company has rules about office timing, deadlines, discipline. Law protects both employer and employee. If employees break laws, they are punished; if they follow rules, they should be rewarded (bonus, appreciation).
Law: Enforced by authority**.** If you break rules, you can be penalized or face action
Ethics: Personal behavior**.** If you act poorly, people form a bad impression, but you may not be punished
Morality (part of ethics) examples: "How are you?" "You look great in this dress." "You were absent yesterday, everything okay?"

Week 2 : Professional Ethics & Code of Ethics

Law: Laws are **rules created by a company or society** that tell people what they **must do** or what they **must not do**. Laws help maintain order and control behavior.
Moral Values: Moral values are **principles of right and wrong**. They guide a person's decisions and behavior. Example: Honesty, kindness, respect.

Ethics: Ethics means **socially acceptable behavior**. Example: Respect for employees, respect for customers.

- **Code of Ethics:** A Code of Ethics explains the **responsibilities of professionals** to act ethically. Applies to employees, companies, and society. Organizations create and distribute these policies and expect workers to follow them.
- **Goals of Code of Ethics** Helps people make good decisions. Includes: Intellectual Property, Privacy, Confidentiality, Professional Quality, Fairness or Discrimination, Liability (Responsibility), Software Risks, Conflicts of Interest, Unauthorized Access to Computer Systems

Detailed Goals of Code of Ethics

1. **Intellectual Property:** Related to creations of the mind. Examples: Software, algorithms, flowcharts, programs, DFDs, research papers, books, patents.
2. **Privacy:** Do not interfere in other people's matters. Example: If managers are in a meeting, don't enter or listen to their discussion.
3. **Confidentiality:** Keep official matters **secret**, do not talk about them publicly. Example 1: After a meeting, keep the termination list confidential. Example 2: System administrators must keep user IDs and passwords secret.
4. **Professional Quality:** Your work should be of **good quality**. People should not complain about it.
5. **Fairness or Discrimination:** Treat everyone equally without bias. Example: A manager should promote the employee who deserves it, not the one who is a personal favorite.
6. **Liability (Responsibility):** Take full responsibility for the tasks assigned to you. Be an **asset**, not a liability for your company.
7. **Software Risks:** Every software project has risks in: Analysis**,** Design**,** Development**,** Implementation
   Examples: Requirements changing (analysis stage) Not enough resources (implementation stage)
8. **Conflicts of Interest:** People can have different opinions. Use **logic and reasoning** to solve conflicts.
9. **Unauthorized Access:** Accessing systems without permission is **unethical and a crime**. Every employee should access only their own computer or system using their user ID and password.

ACM (**Association for Computing Machinery**)**:** World's largest educational and scientific computing society. Brings together computing educators, researchers, and professionals. Goal: Share resources, discuss issues, and solve challenges in computing.

- **ACM Code of Ethics – General Moral Values**
  ACM members must: Contribute to society and well-being, Avoid harming others, Be honest and trustworthy, Be fair; avoid discrimination, Respect copyrights, agreements, and property rights, Respect intellectual property, Respect others' privacy, Maintain confidentiality
- **ACM Code of Ethics – Professional Responsibilities**
  A computing professional must: Try to achieve **highest quality** in work, Improve processes and products, Know and follow professional laws, Give complete evaluation of system risks, Honor contracts and responsibilities, Access systems only when authorized, Obtain and maintain professional competence
- **ACM Code of Ethics – Organizational Leadership**
  A leader must: Explain social responsibilities to employees. Manage people and resources to create quality systems. Ensure computing resources are used properly. Ensure developed systems meet user requirements. Make sure manpower is properly used. Arrange training for employees working on computer systems

Ethical Decision-Making Cases

Case 1 Situation: Alisha, a network manager, is asked to install a cheaper but less secure system. She believes the data is too sensitive for a low-security product.

Questions & Answers

Q1: Which code applies? **A1:** Security and Unauthorized Access

Q2: What should Alisha do? **A2:** (i) Convince management to buy a secure system. (ii) If they refuse, she should resign

Case 2 Situation: Ahmed, a programmer, used programs from 3 teammates but didn't acknowledge them.
He also added code from his friend Rashid but didn't inform management.

Q1: Which code applies? **A1:** Intellectual Property

Q2: Did Ahmed follow ethics? **A2:** No, because: (i) He did not appreciate team members (ii) He violated copyright by using Rashid's code secretly

Case 3 Situation: A software house in Canada delivered a tax software with errors. Clients paid wrong taxes and were penalized.

Q1: Which code applies? **A1:** Violated ACM code – did not honor contracts and responsibilities

Q2: Software house mistakes: **A2:** (i) They did not test software properly. (ii) They made false agreements with clients

Case 4 Situation: Management tells a software manager to hire only white people and prefer females over males.

Q1: Which code applies? **A1:** Fairness and Discrimination

Q2: What should you do? **A2:** (i) Explain disadvantages of discrimination (ii) If management insists, refuse to take part in hiring

Week 3: Software uses, Misuses, Software Development risks

Computer Program: A computer program is a set of instructions written in a programming language for a computer to execute. It is one part of software. Programs are ordered and sequential instructions. Example languages: C++, Java, Python.

Computer Software: Software is a **set of instructions, data, or programs** that help computers run and perform tasks.

Uses of Software: Produces results very fast. Produces accurate results without errors**.** Helps prevent fraud. Helps store and retrieve data from databases**.** Increases employee productivity Examples: Microsoft Word, Excel, PowerPoint**,** Human Resource Management**,** Payroll systems**,** Inventory control**,** Financial accounting**,** Production control**,** Library systems

Misuses of Software: Using software without permission or pirated software violates the copyright act. Software is intellectual property of the creator or company. To use someone else's software legally, a license agreement must be signed.

Software Development Risks: Software development has **internal and external risks**. Identifying risks is **important for quality** and achieving project goals. Developers must **understand risks** to respond effectively.

List of Software Development Risks: Code issues**,** Aggressive deadlines**,** Inaccurate estimations**,** Low productivity**,** Budget issues**,** Poor risk management**,** Inadequate project management**,** Change in project scope**,** Stakeholder issues**,** User response**,** Project team members leaving**,** External risks

Week 4 - Cyber Crime & Intellectual Property

CYBER CRIME: Any criminal activity in which a computer, mobile phone, tablet, or even a landline is used as the main tool.

Forms of Cyber Crime

1. **Fraud**: A wrongful or criminal act done to gain money or personal benefit. *Examples:* A bank employee transfers staff salaries into his own account and escapes. A cashier collects customer cash but does not enter it into the POS system.
2. **Embezzlement**: When someone responsible for handling money accepts payments but does not record the transaction and keeps the money. *Example:* Cashier takes money but does not enter it in the POS machine.
3. **Software/Information Sabotage**: Intentional attempt to damage or corrupt software or data. *Examples:* Infecting a website. Sending viruses targeting MBR or EXE files
4. **Information (Data) Theft**: Illegal transfer or storage of confidential, personal, or financial information. *Examples:* Stealing someone's ID, password, or keys. Unauthorized access to personal accounts
5. **Hacking and Cracking**: Unauthorized access to computer systems or networks.
6. **Digital Forgery (Digital Tampering)**: Manipulating digital documents or images for financial, political, or social gain.

Hacking: Attempt to access a computer or network without permission. Hackers may aim to gather information for illegal purposes. Hackers are intelligent programmers who use their skills negatively. Hacking is done using software/programs.

Cracking: Unauthorized access to steal, corrupt, or view data illegally. Crackers are *not* programmers — they use techniques like guessing PINs or passwords, breaking user IDs, etc.

Difference Between Hacking & Cracking

Hacking	Cracking
Access systems; may or may not cause harm	Access systems *to steal, corrupt, or view data*
Hackers are programmers	Crackers are not programmers
May be curiosity-based	Always illegal and harmful

Digital Forgery: Manipulating digital documents, images, or files. Used for financial, social, or political advantage. Gives misleading information about a product or service.
Security Measures: Track and record user access with servers. Assign unique ID and password to every user. Passwords must include letters, numbers, and special characters. Users should change passwords regularly. Only authorized people should be allowed in sensitive areas like server rooms. Maintain log files with details such as date, time-in/out, name, and purpose of entry.

INTELLECTUAL PROPERTY (IP): Creations of the human mind — creative or intelligent — such as art, designs, music, literature, symbols, programs, images, etc.
Examples of Intellectual Property: Painting (Art), Building design, Novel (literary work), New song (music), Computer program, algorithm, flowchart, Research papers, Symbols (e.g., red signal = STOP), Images and scenery

Types of Intellectual Property:
1. **Copyright:** Protects original creative works. Prevents others from copying or producing similar items. Applies to: Books, Journals, magazines, bulletins, Songs and music, Dramas, theatre productions, Movies, TV programs, talk shows. Pakistan Copyright Act: Copyright Ordinance, 1962 (amended in 2000)
2. **Trademark:** Unique symbol or sign that identifies a product or company. Example: McDonald's logo. Represents the identity of the product. Illegally copying a trademark is punishable by law.
3. **Patent:** Government authorization that gives exclusive rights to make, use, or sell a product for a fixed time. Only the approved party can produce or sell the item. Examples: *Jam-e-Shirin* – Qarshi, *Rooh Afza* – Hamdard Pakistan, *Student Biryani*, *Mazaidar Haleem*
4. **Trade Secret:** Secret formulas or processes that give a company an advantage. If kept private, the protection can last forever. Example: Coca-Cola formula — never patented, kept secret for decades.

Week 5 Cyber Law and Intellectual Property Law

Intellectual Property Rights (IPR)
- Intellectual property and the rights related to it are often **the most valuable assets** for any company, organization, or institution.
- Things that must be protected include: All **information saved on computers or electronic devices** All **paperwork and documents** like: Computer programs, User manuals, Information on websites
- Unauthorized use of intellectual property must be **stopped through law**, and damages should be recovered.
- Since technology keeps upgrading, the law also keeps changing.
- To handle this, **GATT (General Agreement on Tariffs and Trade)** was created by 23 founding members.
- GATT:
    - Covers **intellectual property rights**
    - Is the only **multilateral and legally binding** agreement
    - Is a **legal agreement between countries and companies**
    - Covers **90% of world trade**, although some countries still haven't joined

Does GATT Cover Intellectual Property?
- GATT’s purpose: Promote international trade by **removing trade barriers** like tariffs and quotas.
- Article XX(d) of GATT 1947 / 1994 mentions **intellectual property rights**.
- In Pakistan, the following laws support and follow GATT: Copyright Ordinance 1962, Patent Ordinance 2000, Trade Marks Ordinance 2001

Confidential Information (Covered Under GATT)
- Confidential information = information that is **not public** and **should not be shown to the public**.

- Examples: Personal information (e.g., email password). Trade secrets. Sensitive government information. Work handled by computer scientists like: Algorithms, Computer programs. Any information that must be protected from unauthorized access. **Non-confidential information** can be used by anyone.

Plagiarism: Taking someone else's work or ideas and presenting them as your own. The following actions are counted as plagiarism: Claiming someone's work as your own. Copying words or ideas without credit or reference. Using someone's work without citation

Cybercrime: Any crime involving a **computer or computer network**. A computer may: Be used to commit the crime Or be the target of the crime . Cybercrime can harm: Hardware, Software, Data, Finances
Types of Cybercrime include: Fraud, Embezzlement, Software/Information Sabotage, Data Theft, Hacking & Cracking, Digital Forgery

Week 6- Software House & Its Goals

Software House: A software house is a company that mainly focuses on developing and releasing software products. It creates software for businesses or general consumers, depending on what their clients need. Their main job is to develop, maintain, and distribute software solutions.

Goals of a Software House: Software houses work for their clients using different technologies and platforms such as: PC and laptop software, Mobile applications, Online applications, Web development, Data science algorithms, APIs, etc.
Clients may include large, medium, and small organizations, and each client's needs depend on their business nature and company size.

Principal Goals of a Software Development Team A software development team aims to: Build high-quality software**,** Meet client requirements**,** Use efficient development processes**,** Improve and update products continuously**,** Deliver reliable and scalable software

Types of Software Houses

1. **Product-Based Software Houses:** These companies create their own products and sell them to customers. They keep improving and upgrading their products to stay competitive. They provide ready-made (out-of-the-box) solutions. Examples: Microsoft, Google, HP, Facebook, Adobe, Cisco.
2. **Service-Based Software Houses:** They **build custom software** based on what the client needs. They provide **consulting services** to help organizations choose the right software solutions. They may also provide dedicated development teams to work directly with clients. **Examples of service providers:** TCS, FedEx, InfoSys, Wipro, etc. **Examples of custom software they build:** Inventory control system**,** HR system**,** Payroll system